PREPARED BY:

UNIGLOBAL CONSULTANCY

ENTERPRISE EDITION 2026

BH Kingdom of Bahrain & GCC Regional Compliance Standards

Gulf Enterprise HR & Payroll Solution

Comprehensive System Functionality Specification, Statutory Rules Engine, Hardware Integration Architecture & Operational Workflow Documentation.

Bahrain Labor Law No. 36/2012
Automated GOSI social insurance tiers, LMRA monthly levy, overtime multipliers (1.25x/1.50x/2.00x), and statutory EOSB gratuity formulas.

Smart Card Hardware Bridge
Instant Bahrain CPR chip reading for 2025/2026 back-chip & legacy front-chip cards via SaNDSLabCardBridge on Port 5050.

Executive Laser Print Engines
Official A5 Laser Payslips with dual authorization signatures and 1-31 Landscape Attendance Matrix sheets.

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PRIMARY CONSULTANT
Uniglobal Consultancy
Strategic HR & Business Advisory

TECHNOLOGY PARTNER
SaNDS Lab Middle East W.L.L.
Software Architecture & Automation

Powered by SaNDS Lab Middle East W.L.L.
Salmabad, Kingdom of Bahrain • www.sandslab.com • +97335078079

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Document Governance & Control

Project Name:	**Gulf Enterprise HR & Payroll Solution**
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Compliance Baseline & Standards

Labor Regulation:	Bahrain Labor Law for the Private Sector (Law No. 36/2012)
Social Insurance:	General Organisation for Social Insurance (GOSI Bahrain)
Labor Market:	Labour Market Regulatory Authority (LMRA) Expat Levy
Smart Card Spec:	Bahrain Information & eGovernment Authority (IGA / CIO)
Currency Standard:	ISO 4217 BHD (3 Decimal Places, e.g. BD 1,250.000)
Timezone:	Asia/Bahrain (GMT+3) Standard Arabian Time

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01 Executive Overview & GCC Compliance Framework

1.1 Strategic Solution Purpose

The **Gulf Enterprise HR & Payroll Solution** is a mission-critical Human Resource Management, Workforce Attendance, and Payroll Processing platform engineered specifically for corporate enterprises, contracting companies, retail groups, and service organizations operating in the **Kingdom of Bahrain and the greater Gulf Cooperation Council (GCC) region**.

1.2 Kingdom of Bahrain Statutory Labor Law Compliance (Law No. 36 of 2012)

GOSI Social Insurance

Bahraini / GCC Citizens: 7.00% Employee Share and 12.00% Employer Share on Basic + Housing.
Expatriate Workforce: 1.00% Employee Share and 3.00% Employer Share on Basic Salary.

LMRA Expatriate Levy

Automated accounting for the monthly **10.000 BHD** Labour Market Regulatory Authority (LMRA) work permit levy per expatriate staff member.

Overtime Multipliers (Art. 54)

Normal Day OT: 1.25x hourly wage.
Night (19:00 - 07:00) / Weekend OT: 1.50x.
Official Holiday OT: 2.00x hourly wage.

1.3 Multi-Country GCC Configuration & Regional Presets

The architecture features a **1-Click Regional Preset Engine** that dynamically reconfigures currency codes, symbol displays, decimal precision, social security tiers, and statutory rules for all GCC member states:

Country	Currency	Precision	Social Insurance Agency	Statutory Gratuity Rule	Standard Workweek
Kingdom of Bahrain BH	BHD (BD)	**3 Decimals (0.000)**	GOSI Bahrain (7% / 12% Cit., 1% / 3% Exp.)	15 days (Y1-3), 30 days (Y4+)	Sun – Thu (Fri/Sat Off)
Kingdom of Saudi Arabia SA	SAR (SR)	2 Decimals (0.00)	GOSI KSA (9.75% / 11.75% Saudi)	15 days (Y1-5), 30 days (Y6+)	Sun – Thu (Fri/Sat Off)
United Arab Emirates AE	AED (Dhs)	2 Decimals (0.00)	GPSSA UAE (5% / 12.5% Emirati)	21 days (Y1-5), 30 days (Y6+)	Mon – Fri (Sat/Sun Off)
State of Kuwait KW	KWD (KD)	**3 Decimals (0.000)**	PIFSS Kuwait (10.5% / 11.5% Kuwaiti)	15 days (Y1-5), 30 days (Y6+)	Sun – Thu (Fri/Sat Off)
State of Qatar QA	QAR (QR)	2 Decimals (0.00)	GRSIA Qatar (5% / 10% Qatari)	21 days per year of service	Sun – Thu (Fri/Sat Off)
Sultanate of Oman OM	OMR (RO)	**3 Decimals (0.000)**	PASI / SPF Oman (7% / 11.5% Omani)	15 days (Y1-3), 30 days (Y4+)	Sun – Thu (Fri/Sat Off)

1.4 Currency, Precision & Regional Timezone Standards

In Bahrain, Kuwait, and Oman, currencies are denominated in 1,000 fils/baisa units requiring exact 3-decimal precision (e.g., `BD 1,250.750`). The system guarantees zero rounding discrepancies across basic salaries, variable allowances, hourly overtime fractions, loan EMI deductions, and net payouts. System timestamps and attendance clocks strictly utilize `Asia/Bahrain` (GMT+3) timezone synchronization.

02 System Architecture & Technical Stack

2.1 Pure Core PHP MVC Architecture (Zero-Bloat Foundation)

Unlike conventional bloated enterprise suites that bundle thousands of unnecessary vendor dependencies and suffer from slow execution, this system is engineered with a custom, ultra-lightweight **Pure MVC Core PHP** backend:

- **Zero Heavy Framework Dependencies:** Built with native PHP 8.x eliminating bulky vendor folders, security vulnerabilities in unvetted third-party packages, and memory overhead.
- **Sub-50ms Response Latency:** Clean, compiled routing table and direct PDO singleton connection ensure lightning-fast API responses even under concurrent multi-user load.
- **Stateless RESTful JSON Design:** Clean separation between backend services and client interfaces. Every endpoint operates as a pure REST API supporting `Authorization: Bearer <token>` headers.

2.2 React 19 + Vite Modern Frontend

The user experience is delivered through a high-performance Single Page Application (SPA) built with **React 19** and bundled via **Vite**:

- **Executive Gulf Theme:** Premium corporate aesthetic featuring an Indigo & Slate background with vibrant Blue & Pink accenting, glassmorphic cards, and high-readability typography.
- **Horizontal Top Navigation:** Optimized for wide executive desktop displays and workstations, keeping all 10 core modules accessible in a single top bar.
- **Native Laser Print Engines:** Embedded CSS print drivers specifically formatted for A5 Laser Payslips and 1-31 Landscape Attendance Matrix sheets without third-party PDF pop-ups.

2.3 Relational Database Schema (18 Normalized MySQL Tables)

Core Workforce Tables

`users`	Admin, HR, and ESS login credentials with Bcrypt hashes.
`departments`	Department master records with division mapping.
`designations`	Job titles and hierarchy classifications.
`employees`	Comprehensive 360° master employee records (CPR, passport, etc.).
`employee_documents`	Attachment vault for digital copies with expiry dates.
`settings`	Regional parameters, GOSI rates, overtime multipliers.

Attendance, Shift & Payroll Tables

`shifts` & `shift_roster`	Shift definitions, start/end times, and rotational calendar.
`employee_shift_assignments`	Default and employee-specific shift mappings.
`attendance`	Daily punch registers, biometric CSV logs, overtime hours.
`payroll_periods` & `payslips`	Monthly payroll cycles, processing dates, itemized payslips.
`loans` & `loan_repayments`	Active loans, EMI schedules, repayment ledgers.
`leave_types` & `leave_applications`	Leave entitlements, balances, and multi-tier approval logs.
`indemnity_calculations`	Statutory EOSB gratuity computations and settlement certs.
`holidays`	Gazetted national and official public holiday calendar.

2.4 Phase 2 Native Mobile Application Readiness

Because all backend endpoints adhere strictly to stateless RESTful JSON conventions with token-based authentication, the system is architected for immediate Phase 2 plug-and-play connectivity with **React Native (iOS & Android)** mobile applications for Employee Self-Service (ESS), mobile check-in, leave submissions, and digital payslip viewing without requiring any backend modifications.

03 Role-Based Access Control & User Permissions

3.1 Role Hierarchy & Segregation of Duties

Security governance enforces strict Segregation of Duties (SoD) through three standardized corporate access roles:

ADMIN System Administrator
Full administrative sovereignty. Manages organization-wide settings, GCC country presets, GOSI rates, overtime multipliers, user accounts, role delegations, and database maintenance.

HR ADMIN HR Operations Admin
Day-to-day HR operational authority. Manages employee onboarding, CPR smart card reading, document vault, shift scheduling, biometric attendance imports, payroll processing, and approvals.

ESS Employee Self-Service
Personal portal for individual staff members. View personal profile, submit leave requests, view loan EMI schedules, download monthly A5 payslips, and monitor document expiry countdowns.

3.2 Granular Permissions Matrix Across Core Modules

System Module	System Administrator	HR Operations Admin	Employee Self-Service
Executive Dashboard	Full GCC Radar, Financials & KPIs	Workforce Metrics & Expiry Radar	Personal Attendance & Leave Balances
Employee Management	Full CRUD + Salary Setup	Full CRUD + Smart Card Reader	Read-Only Personal Profile
Document Expiry Vault	View, Upload, Delete Any Document	View, Upload, Delete Any Document	View & Upload Personal Docs
Smart Card CPR Bridge	Full Access (Port 5050 Bridge)	Full Access (Port 5050 Bridge)	No Access
Shift Master & Rostering	Create, Edit, Delete Shifts & Roster	Assign Shifts & Manage Roster	View Assigned Shift Schedule
Biometric Attendance	Import CSV, Edit Logs, Override OT	Import CSV, Edit Logs, Override OT	Web Punch (If Enabled) & Log View
1-31 Matrix Sheet	View All, Department Filter, Print	View All, Department Filter, Print	View Personal Monthly Timesheet
Payroll Calculation Engine	Process, Finalize, Recalculate	Process, Finalize, Print Payslips	No Access
A5 Laser Payslips	Generate, Print All, Sign Off	Generate, Print All, Sign Off	View & Print Personal Payslip
Indemnity & EOSB Engine	Calculate, Override, Final Settlement	Calculate & Print Settlement Cert.	View Estimated Accrued Gratuity
Leaves & Vacations	Approve, Reject, Adjust Balances	Approve, Reject, Review Calendar	Submit Request, View Balance
Loans & Advances	Disburse, Modify, Write-Off	Disburse & Track EMI Deductions	Apply & View Repayment Schedule
Regional Settings	Full Access (Switch GCC Presets)	Read-Only View	No Access

3.3 Authentication & Session Security Standards

Passwords are encrypted using **Bcrypt (PHP `password_hash` with cost factor 12)**. All API communications require a signed Bearer Token transmitted in the HTTP Authorization header. Cross-Origin Resource Sharing (CORS) is strictly regulated, SQL injection is completely prevented through parameterized PDO prepared statements, and all HTML inputs are scrubbed against Cross-Site Scripting (XSS).

04 Executive Dashboard & GCC KPI Radar

4.1 Operational Overview & Flight Deck Architecture

The Dashboard serves as the central flight deck for executive management and HR leaders, providing real-time operational visibility across workforce attendance, payroll liability, and statutory compliance deadlines:

Total Headcount

Active Roster

Real-time count of active employees vs on-leave and probationary staff.

Document Expiry Radar

30 / 60 / 90 Days

Automated alerts for CPR, Passport, and Visa documents nearing expiry.

Monthly Payroll Summary

BHD Payout

Live aggregation of gross salaries, GOSI contributions, and net disbursements.

Live Digital Clock

Asia/Bahrain

GMT+3 synchronized digital clock with daily present/absent counters.

4.2 Document Expiry Radar Engine

The document radar scans all employee identification credentials, passports, residence permits, and certifications, grouping them into intuitive priority buckets:

CRITICAL: < 30 DAYS

Immediate action required. Prominently flagged to prevent LMRA expatriate work permit fines or immigration visa overstay penalties.

WARNING: 30 – 60 DAYS

Proactive preparation window. HR officers can initiate embassy passport renewal applications or medical insurance policy endorsements.

SAFE: > 60 DAYS

Compliant status. Displays remaining countdown days ensuring long-term audit readiness for corporate governance.

4.3 Financial Quick-View & Operational Launchers

Monthly Payroll Financial Snapshot

Aggregates monthly disbursement figures directly from the payroll calculation engine:

- **Gross Salary Commitment:** Total contractual basic + allowances.
- **GOSI Employer Liability:** 12% citizen / 3% expat employer share.
- **Net Payout Obligation:** Cash/WPS transfer total for banking clearance.

One-Click Operational Quick Launchers

Direct execution shortcuts for high-frequency HR administration tasks:

- **Onboard New Employee:** Opens CPR reader & master profile modal.
- **Upload Biometric CSV:** Direct link to punch log ingestion tool.
- **Generate Monthly Payroll:** One-click pay period calculation engine.
- **Calculate EOSB Indemnity:** Instant Article 116 final settlement wizard.

05 Employee Lifecycle & Document Expiry Vault

5.1 360° Comprehensive Employee Master Record

The employee profile captures all regulatory and organizational attributes required under Bahrain and GCC employment laws:

Category	Captured Data Attributes & Specifications
Personal & Identity	Employee Code, CPR / National ID, CPR Expiry Date, Passport Number, Passport Expiry, Visa / Residence Permit Number, Visa Expiry, Driving License Number, Driving License Expiry, Full Name (English & Arabic), Date of Birth, Gender, Blood Group, Marital Status, Religion, Nationality.
Employment & Organization	Department, Designation, Date of Joining, Employment Type (Permanent, Contract, Probation), Work Location, Reporting Manager, GOSI Social Insurance Registration Number, LMRA Work Permit Number.
Bahrain Salary Package	Basic Salary + Housing Allowance (HR) + Transport Allowance + Car Allowance + Mobile Allowance + Food Allowance + Other Fixed Allowances = Gross Monthly Salary.
Disbursement & Banking	Payment Mode (WPS Bank Transfer, Cheque, Cash), Bank Name, IBAN (International Bank Account Number), Swift Code.
Emergency Contacts	Primary Next of Kin Name, Relationship, Contact Phone Number, Local Bahrain Residential Address.

5.2 Document Vault & Dynamic Expiry Countdown Badges

Every employee record includes a dedicated digital document vault supporting high-resolution image and PDF attachments. The system continuously evaluates expiry dates against the current system date and assigns real-time visual badges:

CRITICAL (< 30 DAYS)
Urgent renewal required immediately. Triggers red alert badge on dashboard to prevent LMRA or immigration fines.

WARNING (30 – 60 DAYS)
Renewal preparation window. Amber badge allows HR to initiate passport embassy renewals or medical checkups.

VALID (> 60 DAYS)
Compliant status. Green badge showing exact remaining days countdown until next renewal milestone.

5.3 Search, Advanced Filtering & Export Capabilities

The employee directory includes real-time search across Employee Code, Name, CPR, Passport, Department, and Designation, with instant CSV and Excel export for compliance audits, insurance renewals, and corporate reporting.

06 Bahrain Smart Card (CPR) Hardware Bridge

6.1 SaNDSLabCardBridge Architecture & Technology Stack

One of the crowning technical innovations of the platform is the native **Bahrain CPR Smart Card Reader Integration**. Built as a specialized .NET bridge application (`SaNDSLabCardBridge`), it establishes a direct hardware communication pipeline between standard PC/SC smart card readers (e.g., OMNIKEY, Identiv, ACS) and the HR web application:

Bridge Technical Core

- **Runtime:** C# .NET 8 / Framework 4.8 high-speed local daemon.
- **PC/SC Driver Layer:** Native `pcsc-sharp.dll` interfacing with Windows WinSCard subsystem.
- **Official IGA/CIO Assemblies:** `BH.CIO.Smartcard.Bahrain.dll`, `BH.CIO.Smartcard.IDCardManager.dll`, `BH.CIO.Smartcard.Lookup.dll`.
- **Photo Rendering:** `Magick.NET` decoding embedded biometric citizen photo byte streams into web-ready Base64 PNG formats.
- **Local REST Service:** Embedded HTTP server listening on `http://127.0.0.1:5050`.

Dual CPR Card Generation Compatibility

The Kingdom of Bahrain issued updated smart card designs. SaNDSLabCardBridge dynamically detects the card generation:

- **2025/2026 New CPR Cards (Back Chip):** Reads modern secure memory structures on the rear of the card.
- **Legacy CPR Cards (Front Chip):** Backward compatible with earlier contact chip structures.

6.2 Automated One-Click Onboarding Workflow

1. **Insert Bahrain Smart Card into USB Reader**
   The officer inserts the employee's physical CPR card into any connected USB smart card reader. The reader LED indicates active contact.
2. **Click "Read Card" in HR Solution Onboarding Modal**
   The React frontend sends an asynchronous HTTP GET request to `http://127.0.0.1:5050/api/operation/ReadCard`.
3. **Decryption & Attribute Parsing via SaNDSLabCardBridge**
   The bridge queries the card chip, parses BER-TLV elementary files (EF), maps lookup tables, and decompresses the card photograph.
4. **Automatic Form Population (15+ Fields Filled in 1.2 Seconds)**
   The onboarding form instantly populates CPR Number, Full English Name, Full Arabic Name, Date of Birth, Gender, Nationality, Passport Number, Passport Expiry, Card Expiry, Local Address, and attaches the official digital photo.

6.3 Bridge Launcher Script & Conflict Resolution

The launcher script (`Start_Bahrain_Card_Bridge.bat`) automatically stops conflicting legacy services (`SCardReadServer`), configures Windows URL ACL permissions on Port 5050, and starts the bridge with diagnostics.

07 Shift Management & Multi-Shift Rostering

7.1 Shift Master Configuration Parameters

Enterprises operate diverse operational shifts across corporate, retail, security, and industrial divisions. The Shift Master module allows administrators to define custom shift rules:

Shift Code	Shift Name	Timing Range	Standard Hrs	Grace Period	Night OT Premium
`MS01`	Morning / General Shift	08:00 AM – 04:00 PM	8.00 Hours	15 mins in / 15 mins out	No (Standard Rate 1.25x)
`ES02`	Evening Shift	04:00 PM – 12:00 AM	8.00 Hours	15 mins in / 15 mins out	Partial (After 19:00 -> 1.50x)
`NS03`	Night Shift (Overnight)	12:00 AM – 08:00 AM	8.00 Hours	15 mins in / 15 mins out	**Yes (Full 1.50x Night Rate)**
`SS04`	Split Shift	09:00–13:00 & 17:00–21:00	8.00 Hours	10 mins in / 10 mins out	Partial (Evening segment)
`FL05`	Flexible / Custom Shift	Configurable Start/End	Variable	Configurable	Based on worked hours

7.2 Shift Rostering & Rest Day Scheduling

Shifts can be assigned through a multi-tier hierarchy:

Shift Assignment Hierarchy

- **Department Default Shift:** Default operational shift applied to all employees in a department (e.g. Administration -> Morning Shift).
- **Employee Default Shift:** Overrides department default for specific staff (e.g. Night Security Guard -> Night Shift).
- **Roster Override:** Day-specific assignments from the monthly shift calendar overriding default rules.

Rest Day & Off-Day Detection

The monthly roster allows managers to assign scheduled rest days (`is_off_day = 1`):

- When an employee punches in on an assigned off-day, work is automatically flagged as **Rest Day Work**.
- Automatically applies **1.50x Weekend/Rest Day Overtime Multiplier** under Bahrain Labor Law.

7.3 Night Shift Differential Accounting (Article 54)

Bahrain Labor Law mandates enhanced compensation for nighttime work. When a shift is designated with `is_night_shift = 1`, or when hours fall between 19:00 (7:00 PM) and 07:00 (7:00 AM), the overtime calculation engine automatically calculates the 1.50x night premium differential.

08 Biometric Attendance & Overtime Engine

8.1 Multi-Channel Attendance Ingestion

Workforce attendance is captured through three flexible operational channels:

- **Universal Biometric CSV / Excel Punch Importer:** Ingest punch logs exported from ZKTeco, Suprema, Hikvision, or Anviz biometric devices via drag-and-drop CSV or copy-paste text parser (`EmpCode, Date, CheckIn, CheckOut, Status, OvertimeHours`).
- **Live Web Check-In / Punch Register:** Real-time digital clock marking daily check-in and check-out with automatic late-in tolerance tracking.
- **Manual Attendance Register:** Supervisor override grid for duty exemptions, outdoor duty, or off-site project work.

8.2 Bahrain Statutory Overtime Engine (Law No. 36/2012, Article 54)

The overtime engine automatically evaluates daily attendance logs against the employee's assigned shift and statutory thresholds:

STATUTORY HOURLY WAGE FORMULA (BAHRAIN LABOR LAW)

```
Hourly Wage = (Basic Salary + Fixed Allowances) / (30 Days * 8.0 Standard Hours)
```

Overtime Classification	Statutory Rate	Trigger Condition	Accounting Application
Normal Working Day Overtime	**1.25x (125%)**	Work hours exceeding 8.0 standard shift hours on normal business days.	`OT Pay = Excess Hours * Hourly Wage * 1.25`
Night Overtime Premium	**1.50x (150%)**	Hours worked during nighttime between 19:00 (7:00 PM) and 07:00 (7:00 AM).	`Night OT Pay = Night Hours * Hourly Wage * 1.50`
Weekend / Rest Day Overtime	**1.50x (150%)**	Any work performed on scheduled weekly rest days (Friday / Saturday / Off Day).	`Rest Day Pay = Worked Hours * Hourly Wage * 1.50`
Official Public Holiday Overtime	**2.00x (200%)**	Work performed on gazetted Bahrain national/religious holidays.	`Holiday Pay = Worked Hours * Hourly Wage * 2.00`

8.3 Grace Periods & Half-Day Automation

Configurable late grace tolerance (e.g. 15 minutes) prevents penalizing employees for minor traffic delays. If worked hours fall between 4.0 and 7.0 hours, the day is automatically classified as **Half Day (HD)**, ensuring fair wage adjustments.

09 1-31 Landscape Monthly Attendance Matrix

9.1 Landscape Calendar Grid (Days 1 to 31) Visualization

For executive audit compliance, Ministry of Labour inspection, and payroll validation, the application features an official **1-31 Landscape Monthly Attendance Matrix Sheet Generator**. This module compiles the entire workforce or specific departments into a panoramic monthly calendar view:

Daily Attendance Status Legend

- `P` **Present:** Full day attendance completed within shift hours.
- `A` **Absent:** Unexcused absence. Triggers automatic salary deduction.
- `HD` **Half Day:** Attendance between 4.0 and 7.0 hours.
- `L` **Approved Leave:** Annual, Sick, Casual, or Maternity leave.
- `OFF` **Rest Day / Off:** Weekly scheduled rest day (Friday/Saturday).
- `H` **Public Holiday:** Gazetted Bahrain national holiday.

Audit Aggregation Columns

Each employee row concludes with four automated reconciliation totals:

- **Total Worked Days:** Count of valid working days credited.
- **Total Absent Days:** Deductible unapproved days count.
- **Total Late Marks:** Late arrivals beyond grace tolerance.
- **Total Overtime Hours:** Exact sum of regular, night, and holiday OT hours feeding into the payroll engine.

9.2 Executive Laser Print Engine (A4 Landscape)

The matrix sheet is paired with a dedicated CSS print engine formatted specifically for **A4 Landscape Laser Printing**:

- **Compact Grid Typography:** High-density tabular layout ensuring all 31 days and audit summaries fit seamlessly without horizontal clipping.
- **Departmental Grouping & Filtering:** Export or print all company personnel at once, or filter specifically by Operations, Administration, Engineering, Logistics, or Sales.
- **Formal Verification Signatures:** Each printed matrix includes authorization blocks for *HR Supervisor Verification* and *General Management Approval*.

9.3 Shift Code & Roster Correlation

The matrix displays the employee's assigned shift code (e.g. `MS01`, `NS03`) directly in the employee header row, allowing auditors to immediately verify whether overtime was correctly calculated against standard shift timings.

10 Gulf Payroll Engine & Statutory Deductions

10.1 Batch Payroll Processing Pipeline

The payroll engine processes monthly disbursements through a secure, high-speed automated pipeline. HR administrators select the target Pay Period (Month and Year), and can calculate payroll for **All Staff**, **By Department**, or for an **Individual Employee** in a single click:

CORE NET SALARY CALCULATION FORMULA

```
Net Salary = (Basic Salary + Fixed Allowances + OT Earnings + Bonus) - (GOSI Employee Share + Loan EMI + Absent Deductions)
```

10.2 Social Insurance (GOSI) Statutory Calculations

Workforce Category	Employee Deduction (%)	Employer Contribution (%)	Salary Calculation Base
Bahraini / GCC Citizens BH	**7.00%**	**12.00%**	Basic Salary + Housing Allowance (HR)
Expatriate Workforce 🌎	**1.00%**	**3.00%**	Basic Salary Only

10.3 LMRA Levy, Loan Deductions & Absence Proration

- **LMRA Monthly Expatriate Levy:** Fixed 10.000 BHD monthly employer levy tracked per active expatriate staff member.
- **Automated Loan EMI Deductions:** Active employee loans automatically inject monthly installment deductions, updating outstanding loan ledgers.
- **Statutory Absence Proration:** `Absence Deduction = (Basic Salary / 30 Days) * Absent Days`.
- **Wage Protection System (WPS) Readiness:** Generates official SIF (Salary Information File) structures compliant with Central Bank of Bahrain (CBB) payroll file formats.

10.4 Recalculation & Pay Period Finalization

Draft payroll runs can be recalculated multiple times to account for retroactive overtime adjustments or approved leaves before final approval and payslip generation.

11 Official A5 Laser Payslip Generation

11.1 Industry Standard A5 Laser & Thermal Formatting

In Bahrain and the GCC, the standard medium for confidential salary disbursement is the **A5 Landscape / Portrait Laser Slip**. The system includes an executive laser-optimized payslip generator designed to print two complete payslips per A4 sheet or output directly to dedicated A5 laser/thermal printers:

Dual-Column Financial Breakdown

Clean separation between contractual earnings and statutory deductions:

- **Earnings Column:** Basic Salary, Housing Allowance, Transport, Car, Mobile, Food, Overtime Pay, Bonus, Incentives.
- **Deductions Column:** Employee GOSI (7% / 1%), Active Loan EMI, Unpaid Absence Deductions.
- **Net Payout:** Clear display in figures (BD 1,250.000) and full alphabetical word representation.

Dual Authorization Signature Blocks

Each generated payslip features two official acknowledgment boxes:

- **Prepared & Verified By:** HR & Accounts Officer seal and authorization signature.
- **Employee Acknowledgment:** Formal signature line confirming receipt of net salary without dispute.
- **WPS Compliant:** Includes Bank Name and IBAN for seamless Wage Protection System auditing.

11.2 Single-Click Bulk Printing & Employee Self-Service Delivery

Administrators can batch print all payslips for an entire department or company with a single click. Simultaneously, individual payslips are instantly published to the **Employee Self-Service (ESS)** portal, where staff members can view and download encrypted PDF copies securely.

11.3 Security & Anti-Tampering Features

Each payslip contains a cryptographic transaction hash, document generation timestamp, and unique payslip ID matching the internal database record, preventing salary certificate forgery.

12 Indemnity (EOSB) & Final Settlement Engine

12.1 Statutory Bahrain Gratuity Formula (Law No. 36/2012, Article 116)

When an employee concludes their service, the system calculates End of Service Benefits (Indemnity) based on exact continuous service duration:

Tier 1: First 3 Years of Service

15 Days Basic Salary for each year of continuous service during the first 3 years.

`Yearly Rate = Basic Salary * (15 / 30) = 0.5 * Basic Salary`

Tier 2: Beyond 3 Years of Service

30 Days (1 Full Month) Basic Salary for each subsequent year of service.

`Yearly Rate = Basic Salary * (30 / 30) = 1.0 * Basic Salary`

PRO-RATA FRACTIONAL SERVICE INDEMNITY FORMULA

```
Total Indemnity = (Min(Years, 3.0) * 0.5 * Basic) + (Max(0, Years - 3.0) * 1.0 * Basic) + Pro-Rata Days Calculation
```

12.2 Resignation vs Termination Multipliers & Leave Encashment

- **Termination by Employer / Non-Renewal:** 100% full statutory indemnity entitlement.
- **Voluntary Resignation:** Scaled according to Article 116 service brackets (< 3 yrs = 0%, 3-5 yrs = partial, > 5 yrs = 100%).
- **Unused Annual Leave Encashment:** `Leave Salary = (Basic Salary / 30 Days) * Unused Annual Leave Days`.
- **Final Settlement Certificate:** Official legal clearance statement consolidating Indemnity, Leave Salary, Notice Pay, Loan balances, and Net Final Settlement with full employee discharge waiver.

12.3 Legal Discharge & Ministry of Labour Compliance

The generated Final Settlement Certificate contains standard bilingual legal release clauses confirming the employee has received all statutory dues and releases the employer from further civil or labor claims.

13 Leaves & Vacation Lifecycle Management

13.1 GCC Statutory Leave Types & Policies

The leave management engine administers statutory leave entitlements with multi-tier approval workflows:

Leave Category	Statutory Allocation	Remuneration Policy	Labor Law Reference
Annual Leave	**30 Calendar Days / Year**	Full Pay (Accrues at 2.5 days per month)	Article 58 (Law 36/2012)
Sick Leave	**55 Days Total** (Tiered)	15 Days Full Pay • 20 Days Half Pay • 20 Days Unpaid	Article 65 (Law 36/2012)
Maternity Leave	**60 Calendar Days**	Full Pay for female employees	Article 32 (Law 36/2012)
Hajj Pilgrimage Leave	**14 Calendar Days**	Full Pay (Once during entire employment tenure)	Article 67 (Law 36/2012)
Compassionate / Casual	**3 Days**	Full Pay (Marriage, Bereavement)	Article 63 (Law 36/2012)
Unpaid Leave (LWP)	As approved by Employer	Zero Pay (Deducts from monthly payroll)	Mutual Agreement

13.2 Multi-Stage Approval Lifecycle

Approval Stages

- **Step 1: Submission:** Employee requests leave via portal with dates, reason, handover notes, and replacement contact.
- **Step 2: Department Review:** Line manager reviews coverage and operational requirements.
- **Step 3: HR Final Approval:** HR verifies balance and formally approves, updating attendance registers.

Automatic Payroll Integration

Approved leaves directly interface with the monthly payroll engine:

- Paid leaves are credited as valid worked days.
- Unpaid leaves (LWP) automatically trigger proration salary deductions.
- Tiered sick leaves past day 15 trigger 50% basic salary deductions.

14 Employee Loans, Advances & EMI Schedules

14.1 Loan Origination & EMI Scheduling

To assist workforce welfare while maintaining disciplined corporate accounting, HR administrators can record employee emergency loans, vehicle loans, and salary advances:

Loan Origination Parameters

- **Principal Amount:** Disbursed loan sum in local currency (e.g. BD 600.000).
- **Tenure (Months):** Repayment window (e.g. 6, 12, 24 months).
- **Start Period:** Month and year when first EMI deduction begins.
- **Reason / Category:** Emergency medical, education, family travel, housing advance.

Automated Monthly EMI Calculation

EQUATED MONTHLY INSTALLMENT

```
Monthly EMI = Principal Amount / Tenure Months
```

Fractional decimals are handled cleanly to ensure zero rounding slippage across the repayment lifespan.

14.2 Integrated Payroll Deduction & Balance Depletion

During each monthly payroll batch run, the payroll engine automatically queries active loans:

- The monthly EMI is deducted directly from gross earnings as a statutory/voluntary deduction.
- The deduction is logged into `loan_repayments` with the pay period reference.
- The outstanding loan balance decreases in real-time. Once the balance reaches 0.000, the loan automatically flips to **Fully Paid**.

14.3 Early Settlement & Exit Deductions

Employees can make lump-sum early repayments. In the event of contract termination or resignation, any remaining loan balance is automatically pulled into the **Final Settlement Engine** to offset against accrued EOSB Indemnity before net payout.

15 Regional Settings & Organizational Masters

15.1 One-Click GCC Regional Presets

The Settings module empowers administrators to customize corporate governance rules across the GCC:

1-Click Country Switcher

Instantly switches regional rules between Bahrain (Default), Saudi Arabia, UAE, Kuwait, Qatar, and Oman. Automatically re-maps currency symbols and GOSI tiers.

Currency & Decimals

Precision configuration for 3 decimals (BHD, KWD, OMR) vs 2 decimals (SAR, AED, QAR). Enforces exact accounting standards across all ledger tables.

Timezone & Calendar

Synchronized to `Asia/Bahrain` (GMT+3). Sets default workweek schedule (Sunday–Thursday vs Monday–Friday) and standard daily hours.

15.2 Organizational Masters (Departments & Designations)

Departments Master

Manage corporate operational divisions (Operations, Administration, Human Resources, Engineering, Sales, Logistics):

- Assign default department shift codes.
- Track active headcount tallies per department.
- Filter attendance sheets and payroll runs department-wise.

Designations Master

Maintain official job descriptions, classifications, and executive ranks:

- Executive Management, Project Managers, Engineers, Officers, Technicians.
- Correlates with Bahrain LMRA occupation classification codes.

15.3 Gazetted Public Holidays Calendar

Administrators maintain the annual schedule of official gazetted holidays (National Day, Eid Al Fitr, Eid Al Adha, Hijri New Year, Ashoora, Labor Day). Any attendance punches logged on these dates automatically trigger the statutory **2.00x Public Holiday Overtime Multiplier**.

16 End-to-End Operational Workflow Maps

Workflow 1: Smart Card Onboarding

1. **Insert CPR into USB Reader**
   Connect PC/SC reader; insert card.
2. **One-Click Chip Ingestion**
   Click "Read Card" (Port 5050 Bridge). Auto-populates 15+ fields & photo.
3. **Salary & Shift Assignment**
   Set basic + allowances, assign shift, upload contracts to vault.

Workflow 2: Monthly Attendance & OT

1. **Ingest Biometric CSV Logs**
   Upload monthly biometric device punch exports.
2. **Evaluate Shifts & Overtime**
   Compute normal 1.25x, night 1.50x, weekend 1.50x, holiday 2.00x.
3. **1-31 Matrix Sheet Audit**
   Print landscape matrix for departmental manager verification.

Workflow 3: Monthly Payroll & Payout

1. **Select Pay Period & Calculate**
   Pulls basic, allowances, OT pay, GOSI, loan EMIs, absence deductions.
2. **Review & Finalize Payroll Run**
   Verify total bank commitment and GOSI/LMRA employer liabilities.
3. **Print Official A5 Laser Payslips**
   Batch output payslips with dual authorization signatures.

Workflow 4: Exit & Indemnity Settlement

1. **Select Employee & Exit Reason**
   Termination vs Resignation (Article 116 tenure brackets).
2. **Auto-Compute EOSB & Leave Salary**
   15d Y1-3 + 30d Y4+ pro-rata + unused annual leave encashment.
3. **Generate Final Settlement Certificate**
   Legal discharge & waiver document for LMRA/Ministry clearance.

17 Solution Credentials & Vendor Profiles

PREPARED BY:

Uniglobal Consultancy

Strategic Management, Human Resources Advisory & Regional Corporate Implementation

Uniglobal Consultancy is a premier management consulting and human resource advisory firm delivering strategic business transformations, regulatory compliance frameworks, and organizational excellence across the GCC region. Uniglobal specializes in aligning corporate workforce governance with Bahrain Labor Law, LMRA mandates, and GOSI statutory requirements, ensuring operational efficiency and risk mitigation.

HR Process Engineering
Designing compliant organizational structures, job descriptions, and salary benchmarks.

Labor Law Compliance
Auditing indemnity liabilities, contracts, and employment practices under Law 36/2012.

Enterprise Rollout
End-to-end change management, data migration, and workforce onboarding execution.

POWERED BY:

SaNDS Lab Middle East W.L.L.

Enterprise Software Architecture, Gulf Labor Law Systems, Digital Biometrics & Cloud Automation

SaNDS Lab Middle East W.L.L. is an advanced engineering and enterprise software development firm headquartered in the Kingdom of Bahrain. SaNDS Lab pioneers high-performance cloud architectures, custom ERP solutions, statutory payroll engines, and proprietary hardware bridge drivers connecting web applications directly with national identification cards and biometric terminals.

Corporate Headquarters & Operations

Location:	**Salmabad, Kingdom of Bahrain**
Official Portal:	www.sandslab.com ↗
Product Suite:	sandslab.com/products/ ↗

Direct Technical Support & Assistance

WhatsApp:	**+97335078079** (Live Instant Support)
Email:	info@sandslab.com
Core Domain:	Digital Biometrics, PC/SC Card Bridges & Cloud ERP

Official Release & Document Authorization

This specification document represents the verified functional architecture of the Gulf Enterprise HR & Payroll Solution. All features, calculation formulas, and workflows are certified compliant with the laws of the Kingdom of Bahrain.

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